

VTR-1

The Verifiable Track Record Standard

Requirements for proving a trading performance claim without
trusting the issuer or revealing the strategy

VERSION	STATUS	DATE	LICENSE
1.1	Published — FINAL	August 1, 2026	CC BY 4.0

TIMESTAMP

Bitcoin · OpenTimestamps

Prove the edge. Never reveal the strategy.

Mizan Verification, Inc. · a Delaware corporation

Canonical: mizan.market/standard-spec · Contact: muavia@mizan.market

· Foreword

Finance has standards for how returns are *presented*. It has never had a standard for whether they are *true*. For as long as track records have existed, the number has been graded by the person selling it, and every layer of assurance built on top of it (auditors, administrators, databases) has verified only that the number was reported, never that it was real. Trillions of dollars move on that arrangement.

This document specifies its replacement. Under this standard, a performance claim is not believed; it is **checked** by anyone, in milliseconds, against cryptographic commitments made before the evaluation ran, with no trust placed in the issuer at any step. The strategy itself is never revealed. The number of strategies tried is the input every anti-luck correction depends on, and the one every issuer has always been free to understate. Under this standard it becomes a structural property of the proof instead of a claim.

A word on how this standard was tempered. Its reference implementation's first published verdict was a refusal of its own author's flagship strategy. That result was deflated below significance and published as-is, permanently. Its registry lists the author's failures beside every pass. A verification standard that will not lie for its own creator is the only kind fit to judge anyone else's numbers; this property is a design requirement, not a marketing posture, and §6 exists so that it survives the author.

The standard is published under an open license, deliberately. Anyone may implement it without permission (§7). The claims it stakes are falsifiable, and the document tells you how to falsify them. Credentials issued under it are append-only and independently timestamped: the record of who proved what, and when, is already being written, and it cannot be written retroactively. Implementers, verifiers, and critics are equally welcome. The standard's ambition is not that its author wins, but that "trust me" stops being an acceptable answer in systematic finance.

— *Mohammad Muavia, Founder, Mizan Verification, Inc.* · August 2026

STATUS OF THIS DOCUMENT

This is Version 1.1 of VTR-1, the Verifiable Track Record Standard (known colloquially as VTR-1), published August 2026 and stewarded by Mizan Verification, Inc. It is an open specification: anyone may implement, cite, or extend it with attribution (CC BY 4.0). The canonical, continuously served text is maintained at **mizan.market/standard-spec**; this document is the fixed, citable rendering of that text. The specification is versioned under the same governance it defines (§6): future versions supersede but never invalidate credentials issued under prior versions. Corrections are invited and will be published with attribution.

Freeze. Version 1.1 is **frozen**. Its normative text will not change. Findings after this date are published as errata with attribution. Where a finding alters a requirement it is incorporated into a numbered successor version under the governance of §6, which supersedes but never invalidates credentials issued under this one. Requirement identifiers R-01 to R-41 are permanent: a retired requirement keeps its number instead of having it reused, so a conformance claim written against this version stays readable forever. **The digest of this version, and of every superseded rendering of it, is published at mizan.market/standard-spec. A copy whose digest appears in neither list is not this version.**

Version history. 1.0, 31 July 2026: initial publication, sections 1 through 8. 1.0, 1 August 2026: sections 9 through 11 and Appendices A, B and C added before first citation. Section 9 follows the first external submission, which demonstrated that strategy expressiveness in a declared representation is bounded and must be disclosed rather than assumed. Appendix B assigns stable identifiers to every normative requirement so that a conformance claim can be contradicted. Appendix C accompanies the publication of the conformance vectors. No requirement stated on 31 July was weakened or removed. Appendix D pins the estimators and the arithmetic; Appendix E pins the commitment constructions, without which the requirements of §3 and §4 are not independently reproducible. A hostile review of the document on 1 August produced twelve findings, all addressed in the same revision: §5 was **narrowed** so that path-dependent exit logic is permitted under the conservative fill discipline of R-27 while high-frequency strategies remain excluded without exception; the gate policy was **defined** rather than referred to, and this specification does not prescribe threshold values; R-15 was **corrected**, because as first published it misdescribed §5; R-36 to R-41 were added to close requirements that existed in prose without identifiers; and a priority claim was **removed** from §7 as inappropriate to a normative section. The findings and their resolutions are recorded here instead of being silently applied. **1.1, 1 August 2026:** published before any citation of 1.0, which remains available and verifiable. No requirement was added, removed, or altered in meaning, and all 41 identifiers keep their text. Two errata filed against 1.0 are resolved here. The Status block no longer claims that a differently hashing copy is a different version, since 1.0 was itself published in more than one rendering. The plural "authors" in §11 and Appendix C is corrected to the singular. The remainder of the revision is editorial: punctuation and sentence rhythm were reworked for readability, and a requirement-by-requirement comparison of 1.0 and 1.1 is published with the errata.

CONFORMANCE KEYWORDS

The key words **MUST**, **MUST NOT**, **SHALL**, **SHOULD**, **RECOMMENDED**, and **MAY** in this document are to be interpreted as described in RFC 2119 and RFC 8174 when, and only when, they appear in capitals.

ABSTRACT

This document specifies an open standard for verifiable track records: a trading strategy's performance claim, proven honest by cryptographic proof instead of asserted by its author. A conforming credential establishes, to any verifier trusting no issuer, that a strategy's reported statistics were computed correctly, on committed data, net of a committed cost floor, deflated for the committed size of the search that produced it, without revealing the strategy itself. The standard defines three conformance Levels, the exact requirements of each, the verification procedure, the boundaries it refuses to certify, and the governance under which it evolves. It states an adversary model and the attack classes a conforming implementation must address; it distinguishes the two representations a strategy may take and requires that the choice be disclosed; and it pins the estimators, the arithmetic, and the commitment constructions to the level of detail an independent implementation needs to reproduce a credential exactly. Every normative requirement carries a stable identifier so that a conformance claim can be contradicted, and known-answer conformance vectors are published alongside. It is designed to be implemented by parties other than its author, and to be checkable, not believed. The key words **MUST**, **MUST NOT**, **SHALL**, **SHOULD**, and **MAY** are used as normative requirement levels throughout.

HOW TO CITE

Muavia, M. (2026). *VTR-1: The Verifiable Track Record Standard, Version 1.1*. Mizan Verification, Inc. <https://mizan.market/standard-spec>

CONTENTS

1 Scope and Purpose	§1
2 Definitions	§2
3 Conformance Levels — Honest Execution • Deflated for Trials • Sealed & Allocatable	§3
4 Verification Procedure	§4
5 What This Standard Does Not Certify	§5
6 Governance (Era Law)	§6
7 Implementation and Conformance Claims	§7
8 Citation and Attribution	§8

9 Strategy Representation — declared and committed	§9
10 Security Considerations — adversary model and attack classes	§10
11 Prior Work	§11
A Appendix: Conformance Checklist	§A
B Appendix: Index of Normative Requirements — R-01 to R-41	§B
C Appendix: Conformance Vectors — nine known-answer tests	§C
D Appendix: Normative Statistical and Numerical Definitions	§D
E Appendix: Commitment Constructions — SHA-256, leaves, trees, ledger	§E

1 Scope and Purpose

Every track record in systematic finance is, today, graded by the party with the greatest incentive to shade it, and the tools that check it verify only that a number was *reported*, not that it is *true*. This standard exists to close that gap. It defines what it means for a performance claim to be **verifiable**: reproducible by an independent party, from committed inputs, with no trust placed in the issuer.

This standard governs the **claim**, not the strategy. A conforming credential proves properties of a track record while the strategy, its parameters, and (at Level III) its data remain private. It is asset-class agnostic: equities, futures, FX, commodities, crypto, and rates are all in scope where a canonical, committable price series exists.

2 Definitions

Credential.

A cryptographic proof object asserting that a strategy's committed statistics cleared a defined gate on committed data. A credential is **PASS** or **FAIL**; both are valid credentials.

Committed data.

A price series bound to a Merkle root that the proof references; the verifier re-derives the root from its own copy and **MUST** reject on mismatch.

Committed cost floor.

A minimum per-side fee and slippage. The committed costs are charged in-circuit; the floor itself **MUST** be checked by the conforming verifier, and **SHOULD** be enforced in-circuit. An issuer **MAY** commit higher costs but **MUST NOT** commit lower.

Trial ledger.

A commitment to the complete set of strategies evaluated in a search. The **trial count N** is the leaf count of this commitment, not a value the issuer reports.

Gate policy.

The set of thresholds a credential's metrics are tested against. A gate policy **MUST** be pinned by the era, published by the implementation, and identical for every credential of a given Level and era; it **MUST NOT** be selectable by the prover per credential. **This standard does not prescribe threshold values.** Those express an implementation's risk posture, not a property of verifiability. It requires only that whatever policy is used be committed, published, and re-checkable.

Era.

The exact proving program, identified by its cryptographic image hash. A credential names the era that judged it and remains verifiable against that era permanently.

Verifier.

Any independent party running the era-pinned verification procedure. The verifier trusts no issuer and re-derives every asserted property locally.

3 Conformance Levels

A credential **MUST** declare exactly one Level. Higher Levels subsume the requirements of all lower Levels.

LEVEL I Honest Execution

The claim: this performance actually happened, as stated, on real data, net of real costs.

A Level I credential **MUST** prove, in-circuit:

1. The reported return, Sharpe, drawdown, and worst-bar are the correct results of applying the committed strategy to the committed data.
2. The committed per-side costs were charged on every transaction; that they meet or exceed the floor is confirmed by the verifier (\$4, step 5), with in-circuit floor enforcement **RECOMMENDED**.
3. No lookahead: the position at bar t depends only on data through bar t .
4. The annualization basis is consistent with the committed data window (calendar-honest; over-counting is rejected).
5. The strategy specification is **absent** from the proof, present only as a sealed commitment.

Level I answers *did this happen?* It does not address luck.

LEVEL II Deflated for Trials

The claim: this result is not merely real, but unlikely to be luck given how it was found.

A Level II credential **MUST**, in addition to Level I:

1. Commit the complete trial ledger before evaluation, so that **N is the leaf count** of the committed tree and cannot be understated.
2. Force the presented winner, in-circuit, to be the maximum of the committed trials.
3. Recompute the anti-overfitting statistics inside the proof: at minimum the **Deflated Sharpe Ratio** (Bailey & López de Prado) using the committed N; a conforming implementation **SHOULD** also compute the **Probability of Backtest Overfitting** (CSCV) and **MAY** compute the Probabilistic Sharpe Ratio, the Superior Predictive Ability test, and combinatorial purged cross-validation over the committed set.
4. Report every computed statistic, pass or fail. A credential **MUST NOT** report a subset that flatters the result.

Level II answers *is it skill or luck, given the search?* It is the level at which the standard's central contribution, the committed trial ledger, binds.

LEVEL III Sealed & Allocatable

The claim: everything above, provable by a manager who reveals nothing, to an allocator who trusts no one.

A Level III credential **MUST**, in addition to Level II:

1. Prove the track record while the strategy or model runs entirely on the submitter's own machine, submitting only committed decisions (the sealed path). The model **MUST NOT** be recoverable from the credential.
2. Bind an owner key so the credential's holder is provable.
3. Be extensible as a **chained track**: the same committed strategy re-proven over a growing data window, each link cryptographically bound to its parent, so that bars arriving after a commitment carry forward-evidence weight.

Level III is the allocatable credential: a manager proves an edge to an allocator without disclosure, and the allocator verifies it without trust.

4 Verification Procedure

A conforming verifier **MUST**, given a credential and the verifier's own copy of the data:

1. Confirm the proof's guest image hash equals the era the credential names.
2. Verify the cryptographic proof.
3. Re-derive the data root from the verifier's own data and confirm it matches the committed root.
4. Confirm the gate policy committed in the proof is the policy the implementation publishes for that era and Level (\$2), and that it was not selected per credential.
5. Confirm costs meet or exceed the floor and the annualization basis is bound to the data.
6. Confirm the trial count is the committed leaf count (Level II+) and the winner is its maximum.
7. Emit the verdict. The verifier **MUST** return a failure indication on any check that does not pass, and **MUST NOT** require trust in the issuer at any step.

A verification implementation **SHOULD** be distributable and runnable offline. The reference verifier completes in milliseconds and is available under an open license.

5 What This Standard Does Not Certify

Stated normatively, because the boundary is the standard's integrity:

- A credential is a **seal on history**, not a forecast. It **MUST NOT** be represented as predicting future returns.
- It does **not** model capacity, market impact, or fill quality beyond the committed cost floor, and **MUST NOT** be represented as doing so.
- **Fills that the committed data cannot justify are out of scope.** Bar data cannot establish that a trade occurred at a price the bar's range does not contain; a proof of a fill that did not happen is a proof of fiction. A conforming implementation **MUST** either refuse such a strategy, or evaluate it under the conservative fill discipline of R-27 and disclose the assumptions made. Path-dependent exit logic (profit targets, stop levels, time-based exits) is permitted on that condition and on no other.
- **High-frequency strategies are out of scope permanently, without exception,** because their economics live below the resolution of any committable bar series. No fill discipline rescues them, and a conforming implementation **MUST** refuse them.
- The trial ledger binds the *committed* search. Trials run off-ledger are not covered; closing that residual requires forward pre-registration (an anchored ledger timestamped before the out-of-sample bars exist). A credential **MUST** disclose whether it is a sealed backtest or a pre-registered forward record.

6 Governance (Era Law)

The engine that issues credentials evolves; the standard governs that evolution so that no issuer, including the standard's author, can retroactively change what a past credential means.

1. **A credential names its judge.** It commits the image hash of the era that produced it.
2. **Eras are superseded, never revoked.** A changed engine is a new era with a new hash; old credentials verify against their own era permanently.
3. **A changed judge is a new judge.** Any change reaching the compiled proving program is, by definition, a new era.
4. **Weakness is disclosed, not rewritten.** A soundness gap in a frozen era is published as status, with the credential remaining checkable; the reader, not the issuer, decides its worth.

This specification is itself versioned under the same discipline. This is **Version 1.1**; future versions supersede but never invalidate credentials issued under prior versions.

7 Implementation and Conformance Claims

Anyone **MAY** implement this standard. A party claiming conformance **MUST** publish, for each credential: its declared Level, the era hash, the committed data root, and a re-verifiable proof object. A party **MUST NOT** describe a credential as conforming to a Level whose requirements it does not meet.

A reference implementation, a public registry of conforming credentials, including the issuer's own published failures, and an open verifier are maintained at **mizan.market**. Conformance vectors for testing an independent implementation are published under Appendix C. **This specification makes no claim about the relative merit or priority of any implementation, including its steward's.** Conformance is established by the requirements of Appendix B and the vectors of Appendix C. It is tested, not asserted.

8 Citation and Attribution

This standard implements published statistical and cryptographic work; the full citation lineage is maintained at **mizan.market/research/lineage**. None of the cited authors has endorsed this standard; the citations run one direction. This document implements their work, and its author takes responsibility for the implementation.

Corrections to this specification are invited and will be published, with attribution, under the same discipline applied to the reference implementation's own numbers: **muavia@mizan.market**.

Errata. Version 1.0 being frozen, a correction to this text is published as an erratum rather than applied silently to the document. An erratum records what was found, who found it, and what follows: no change where the text is defensible as written, an entry in the errata list where the text is unclear, or a numbered successor version where a requirement genuinely changes. **Errata are published whether or not the finder is named, and whether or not the finding flatters the standard.** Send them to **muavia@mizan.market**.

Reporting a soundness break. A conforming implementation **MUST** publish a route by which a suspected soundness break can be reported, and **MUST** disclose a confirmed break as era status under §6. The affected era is marked, not rewritten, and its credentials remain checkable so that readers can judge them for themselves. For this specification and its reference implementation that route is **muavia@mizan.market**. A report will be acknowledged, and a confirmed break published, whether or not the finder wishes to be named. **An implementation that quietly patches a soundness break has revoked a credential without saying so, which §6 forbids.**

9 Strategy Representation

A conforming implementation **MUST** disclose, on every credential, which representation the evaluated strategy took. Two are defined.

Declared representation.

The strategy is expressed in a specification language the implementation evaluates directly. The representation is legible to anyone holding the specification, but the implementation's language bounds what can be expressed. An implementation **MUST NOT** silently substitute a strategy it can express for one it cannot: if a submitted strategy is not expressible, the implementation **MUST** refuse or disclose the substitution on the credential.

Committed representation.

The strategy is executed by its owner, which commits only its resulting decisions, and the implementation evaluates the committed decision stream. Any strategy is expressible in this form, including path-dependent logic such as profit targets, stop levels and time-based exits. The logic itself is not revealed and therefore not legible from the credential alone.

The two are complementary rather than ranked. A submitter may obtain legibility *and* expressiveness by publishing a deterministic encoder alongside a committed-representation credential, so that a reader verifies strategy to decisions with the encoder and decisions to metrics with the proof. **This specification does not require any implementation to support arbitrary strategy expressiveness in its declared representation, and an implementation MUST NOT imply that it does.**

10 Security Considerations

This section states the adversary model and the attack classes a conforming implementation is expected to address. It is stated normatively because a verification standard that does not name its attacks is decoration.

Adversary model. The adversary is the submitter, and in the limit the issuer. A conforming implementation **MUST** be sound against a submitter who controls their own inputs and is motivated to obtain a favourable verdict, and its governance (§6) **MUST** be sound against an issuer motivated to change what a past credential meant.

Trial-set padding.

Duplicate or near-duplicate candidates inflate N while adding no dispersion, which can collapse a selection-bias correction toward neutrality. A conforming implementation **SHOULD** refuse identical and exactly proportional candidates and **MUST** disclose any residual it does not detect.

Decoy candidates.

Deliberately degenerate entries can flatter a favoured result by comparison. Corrections robust to this class, such as studentized multiple-testing procedures, **SHOULD** be preferred over those that are not.

Annualization inflation.

Overstating periods per year inflates every annualized statistic. An implementation **MUST** bound the declared periodicity against the committed data window (§4, step 5).

Data substitution.

Evaluating on data other than that claimed. Prevented by requiring the verifier to re-derive the data commitment from its own copy (§4, step 3).

Off-ledger search.

Candidates evaluated outside the commitment are invisible to the correction. This is a disclosed residual (§5), not a solved problem; closing it requires forward pre-registration.

Counterfactual fills.

Path-dependent exit logic can book prices the market never traded, for example an exit at a stop level the instrument gapped past. An implementation evaluating such logic **MUST** use conservative fill assumptions and disclose them, and **MUST NOT** assume a fill it cannot justify from the committed data. **A proof of a fill that did not happen is a proof of fiction, and encoding one inside a credential is worse than not offering the feature.**

Issuer capture.

Retroactive alteration of the judging program. Prevented by era law (§6): the credential commits the image of its judge, programs are superseded, never revoked, and archived builds remain

independently runnable.

11 Prior Work

This standard specifies enforcement for statistical corrections it does not originate. The deflation program derives from Bailey and López de Prado, including the Probabilistic Sharpe Ratio (2012), the Deflated Sharpe Ratio (2014), and the Probability of Backtest Overfitting via combinatorially symmetric cross-validation (Bailey, Borwein, López de Prado and Zhu, 2017), together with purged and embargoed combinatorial cross-validation (López de Prado, 2018). The multiple-testing program derives from White's Reality Check (2000), Hansen's test for Superior Predictive Ability (2005), and the multiple-testing haircuts and minimum backtest length of Harvey and Liu (2015). The characterisation of why a researcher cannot reliably count their own trials after the fact is due to Gelman and Loken (2014). The cryptographic constructions rest on Goldwasser, Micali and Rackoff (1985, 1989), Merkle (1987), and the transparent proof systems of Ben-Sasson and collaborators (2018).

None of the cited authors has reviewed or endorsed this specification. The citations run in one direction: this document specifies the enforcement of their published work, and its author takes responsibility for the enforcement.

A Appendix: Conformance Checklist

An implementation claiming conformance **SHOULD** be able to answer each of the following affirmatively, and a party evaluating a conformance claim **MAY** use this list directly.

1. Does every credential name the exact judging program by cryptographic image, and does an archived build of that program still verify it?
2. Can a verifier re-derive the data commitment from a copy of the data obtained independently of the issuer?
3. Is the trial count derived from a commitment made before evaluation, rather than declared?
4. Is the presented winner constrained to be the maximum of the committed set?
5. Are all computed statistics reported, including those that fail?
6. Are costs charged on every transaction, and is the floor checked at verification?
7. Is the annualization basis bounded by the committed data window?
8. Is the strategy representation (§9) disclosed on the credential?
9. Are FAIL credentials issued, published, and verifiable on the same terms as PASS credentials?
10. Are known unaddressed attack classes disclosed rather than omitted?

An implementation that answers ten of ten and publishes its refusals is conforming in substance as well as in form. **An implementation that publishes only its passes has not been evaluated by anyone, including itself.**

B Appendix: Index of Normative Requirements

Every normative requirement in this specification is assigned a stable identifier below. A conformance claim **SHOULD** reference these identifiers, so that a claim is precise and a gap is nameable: "conforms to VTR-1 Level II except R-27" is a meaningful statement, while "conforms to VTR-1" alone is not. Identifiers are stable across revisions of this version; a retired requirement keeps its identifier rather than having the number reused.

R-01	§2	MUST	The verifier re-derives the data root from its own copy of the data and rejects on mismatch.
R-02	§2	MUST	The cost floor is checked by the conforming verifier. It SHOULD additionally be enforced in-circuit.
R-03	§2	MUST NOT	An issuer MAY commit costs above the floor but MUST NOT commit costs below it.
R-04	§3	MUST	A credential declares exactly one conformance Level.
R-05	§3	MUST	A Level I credential proves in-circuit: strategy-to-returns evaluation on the committed data, costs charged on every transaction, and the reported metrics.
R-06	§3	MUST	A Level II credential additionally commits the trial ledger before evaluation, derives N from it, and recomputes the anti-overfitting statistics in-circuit.
R-07	§3	MUST NOT	A credential does not report a subset of results that flatters the outcome.
R-08	§3	MUST	A Level III credential additionally satisfies the pre-registration and forward-record requirements of §3.
R-09	§3	MUST NOT	The strategy or model is not recoverable from the credential.
R-10	§4	MUST	A conforming verifier performs every step of the verification procedure in §4.
R-11	§4	MUST	The verifier returns a failure indication on any check that does not pass.
R-12	§4	MUST NOT	The verifier does not require trust in the issuer at any step.
R-13	§4	SHOULD	A verification implementation is distributable and runnable offline.
R-14	§5	MUST NOT	A credential is not represented as predicting future returns.
R-15	§5	MUST	A conforming implementation refuses a strategy whose result depends on fills the committed data cannot justify, or evaluates it under R-27 and discloses the assumptions made. High-frequency strategies are refused without exception. <i>(Corrected 1 August 2026: as first published, this identifier misdescribed §5.)</i>
R-16	§5	MUST	A credential discloses whether it is a sealed backtest or a pre-registered forward record.
R-17	§7	MAY	Anyone may implement this standard.
R-18	§7	MUST	A party claiming conformance publishes, per credential: declared Level, era hash, committed data root, and a re-verifiable proof.
R-19	§7	MUST NOT	A party does not describe a credential as conforming to a Level whose requirements it does not meet.

R-20	§9	MUST	A conforming implementation discloses, on every credential, which strategy representation was used.
R-21	§9	MUST NOT	An implementation does not silently substitute an expressible strategy for one it cannot express; it refuses, or discloses the substitution.
R-22	§9	MUST NOT	An implementation does not imply arbitrary strategy expressiveness in its declared representation.
R-23	§10	MUST	The implementation is sound against a submitter controlling their own inputs; its governance is sound against an issuer motivated to change what a past credential meant.
R-24	§10	MUST	Any trial-padding residual the implementation does not detect is disclosed. Identical and exactly proportional candidates SHOULD be refused.
R-25	§10	SHOULD	Corrections robust to decoy candidates are preferred over those that are not.
R-26	§10	MUST	The declared annualization basis is bounded against the committed data window.
R-27	§10	MUST	Path-dependent exit logic uses conservative fill assumptions, discloses them, and assumes no fill that cannot be justified from the committed data.
R-28	§D.1	MUST	All committed and re-derived statistics use deterministic fixed-point integer arithmetic at scale 10^6 ; floating point is not used in the evaluated path.
R-29	§D.1	MUST	Verification equality is exact; no numerical tolerance is introduced.
R-30	§D.2	MUST	Population moments, standardised before third and fourth powers; raw kurtosis (3 for a normal); per-period Sharpe.
R-31	§D.3	MUST	PSR is the Bailey & López de Prado estimator as stated, with the radicand floored positive.
R-32	§D.4	MUST	SR_0 is the expected maximum of N trials as stated, evaluated via the log-tail branch for large N.
R-33	§D.5	MUST	Effective sample size under AR(1), one-sided: n_{eff} never exceeds n.
R-34	§D.6	MUST	$DSR = PSR(SR_0)$ at n_{eff} ; MinBTL rounded up, floored at 2.
R-35	§D.7	MUST	Annualisation is a reporting transformation only and does not enter any significance calculation.
R-36	§5	MUST NOT	A credential is not represented as modelling capacity, market impact, or fill quality beyond the committed cost floor.
R-37	§2, §4	MUST	The gate policy is pinned by the era, published by the implementation, identical across credentials of a given Level and era, and not selectable by the prover per credential.
R-38	§D.4	MUST	σ_{SR} is the population standard deviation of the per-period Sharpe across the resolved trials of the committed ledger.
R-39	§D.8	MUST	Where PBO is computed, it uses the CSCV construction of D.8: contiguous equal sub-periods, all balanced partitions, split canonical and not prover-chosen.
R-40	§E	MUST	Commitments use the constructions of Appendix E: SHA-256, domain-separated and index-bound leaves, promotion on odd levels.

R-41	§8	MUST	An implementation publishes a route for reporting a suspected soundness break, and discloses confirmed breaks as era status under §6.
-------------	----	-------------	---

c Appendix: Conformance Vectors

Known-answer tests for this specification are published as **VTR-1 Conformance Vectors v1.0**, a self-contained offline kit of nine vectors: five that **MUST** verify, three that **MUST NOT**, and one published honest failure. Each vector names the requirement identifiers it exercises, and each states a normative exit code so that conformance can be tested automatically rather than asserted.

The negative vectors carry the most weight. One presents a byte-mutated copy of the committed data and **MUST** be rejected (R-01); one presents a sealed credential without its aggregate and **MUST** fail closed rather than degrade to a partial pass (R-11); one presents a credential to the engine of a different era and **MUST** fail on the image identifier, which is the governance model of §6 observable as a test.

The kit is published under CC BY 4.0 and exists for a single purpose: **to make an independent second implementation cheaper to build**. A specification with one implementation is a description of that implementation. The author cannot close that gap alone, and publishing the tests is the part of it that is within his control.

D Appendix: Normative Statistical and Numerical Definitions

Sections 3 and 4 require an implementation to recompute the anti-overfitting statistics. Those requirements are not reproducible unless the estimators and the arithmetic are pinned, because the published literature admits several conventions and two honest implementations choosing differently will disagree on identical inputs. This appendix pins them. **An implementation that follows the cited papers but not this appendix may be statistically defensible and still non-conforming, because it will not reproduce the conformance vectors of Appendix C.**

D.1 Arithmetic

R-28 (MUST). All statistics that a credential commits or a verifier re-derives are computed in **deterministic fixed-point integer arithmetic** with scale $S = 10^6$; a value v is represented by the integer $\text{round}(v \cdot S)$. Floating-point arithmetic **MUST NOT** be used anywhere in the evaluated path. Division truncates toward zero. Intermediate products are computed at 128-bit width before rescaling.

R-29 (MUST). Verification equality is **exact**. There is no numerical tolerance, and a conforming implementation **MUST NOT** introduce one. Because the arithmetic is integer and deterministic, two conforming implementations agree bit-for-bit or one of them is wrong. This is a deliberate departure from ordinary quantitative practice, where a tolerance would be normal: a verifier that accepts "close enough" cannot distinguish a rounding difference from a manipulated statistic.

D.2 Moments of the return series

R-30 (MUST). Given n per-period returns r_i , an implementation computes **population** moments (denominator n , not $n-1$), standardising deviations before taking third and fourth powers:

$$\begin{aligned}\mu &= \frac{1}{n} \sum_i r_i \\ m_2 &= \frac{1}{n} \sum_i (r_i - \mu)^2, \quad \sigma = \sqrt{m_2} \\ z_i &= \frac{r_i - \mu}{\sigma}, \quad \gamma_3 = \frac{1}{n} \sum_i z_i^3, \quad \gamma_4 = \frac{1}{n} \sum_i z_i^4 \\ \hat{S}R &= \frac{\mu}{\sigma}\end{aligned}$$

γ_3 is skewness; γ_4 is the **raw** fourth-moment ratio (3 for a normal, **not** excess kurtosis); the Sharpe is **per-period**, not annualised.

The kurtosis convention is the most common source of silent disagreement between implementations, and the per-period Sharpe convention the second. Both are stated here for that reason. An implementation returns "undefined" rather than a value for $n < 2$ or zero variance.

D.3 Probabilistic Sharpe Ratio

R-31 (MUST). The PSR is the Bailey & López de Prado estimator, evaluated on the moments of D.2:

$$\text{PSR}(\text{SR}_0) = \Phi\left(\frac{(\hat{\text{SR}} - \text{SR}_0) \sqrt{(n-1)}}{\hat{s}}\right)$$

where the denominator is the sampling error of a Sharpe estimated from non-normal returns:

$$\hat{s} = \sqrt{(1 - \gamma_3 \hat{\text{SR}} + \frac{\gamma_4 - 1}{4} \hat{\text{SR}}^2)}$$

The radicand is floored at the smallest positive representable value, so that a pathological moment combination degrades the statistic rather than dividing by zero.

D.4 The deflation benchmark

R-32 (MUST). The benchmark SR_0 is the expected maximum Sharpe of N independent trials, with γ the Euler–Mascheroni constant (0.5772156...) and σ_{SR} the **population** standard deviation (denominator N) of the per-period Sharpe across the **resolved** trials of the committed ledger (unresolved entries are excluded from the dispersion but still counted in N):

$$\text{SR}_0 = \sigma_{\text{SR}} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

For large N the quantity $1/N$ falls below the representable grid and forming $(1 - 1/N)$ loses it entirely. An implementation **MUST** therefore evaluate the upper quantile directly from $-\ln(1/N) = \ln N$ in that regime rather than through the complement. An implementation that does not will silently saturate SR_0 as the trial count grows, producing a benchmark that stops rising exactly when the search gets large enough to matter. That is the failure mode the deflation exists to prevent.

D.5 Serial correlation

R-33 (MUST). The sample length entering the PSR is the effective sample size under AR(1) serial correlation, with ρ the lag-1 autocorrelation of the return series:

$$n_{\text{eff}} = n \frac{1 - \rho^+}{1 + \rho^+}, \quad \rho^+ = \max(0, \rho), \quad \text{floored at } 2$$

The correction is deliberately **one-sided**. Positive autocorrelation shrinks the effective sample, widening the sampling distribution and making the significance test harder. Negative autocorrelation is floored at zero rather than being allowed to inflate n , so the correction can never flatter a result. n_{eff} **never exceeds** n .

D.6 Deflated Sharpe Ratio and Minimum Backtest Length

R-34 (MUST). $\text{DSR} = \text{PSR}(\text{SR}_0)$ evaluated with n_{eff} in place of n , using the moments of D.2, the benchmark of D.4 and the correction of D.5. The Minimum Backtest Length is the smallest sample

length at which the observed per-period Sharpe exceeds the noise ceiling of N trials, $n > (E_{\text{unit}}/\hat{SR})^2$, rounded **up** and floored at 2, against the claimant in both directions.

D.7 Annualisation

R-35 (MUST). Where an annualised figure is reported, the factor is $\sqrt{\text{periods per year}}$ applied to the per-period Sharpe, and the declared periods-per-year **MUST** be bounded against the committed data window per R-26. All statistical tests in this appendix are performed on **per-period** quantities; annualisation is a reporting transformation applied afterwards and **MUST NOT** enter any significance calculation.

D.8 Probability of Backtest Overfitting (CSCV)

R-39 (MUST where computed). PBO is computed by combinatorially symmetric cross-validation (Bailey, Borwein, López de Prado and Zhu). The T observations are partitioned into S **disjoint, contiguous, equal-length** sub-periods. For every **balanced** partition of those sub-periods into a training half and its complement, all $C(S, S/2)$ of them with each partition and its complement both counted, the trials are ranked in-sample, the in-sample best is selected, and the procedure records whether that trial's out-of-sample rank falls below the median. PBO is the fraction of partitions for which it does. Three properties are normative. **The split is canonical, not prover-chosen.** An implementation that lets the submitter influence the sub-period boundaries has not computed PBO, it has computed a number the submitter selected. **Every trial's per-sub-period Sharpe is re-derived from its committed specification on the committed bars**, never declared. **The procedure fails closed:** any binding failure or degeneracy yields $\text{PBO} = 1.0$, which fails any sane threshold, rather than an error the reader might overlook. PBO measures overfitting *of the selection among the committed trials*. It does not establish that the committed set was the whole search; that residual is §10, off-ledger search.

E Appendix: Commitment Constructions

R-01 requires a verifier to re-derive the data root from its own copy of the data, and §2 derives the trial count from the leaf count of a commitment. Neither is reproducible unless the constructions are pinned. This appendix pins them. **Appendix D makes the statistics reproducible; this appendix makes the commitments reproducible. An implementation needs both to pass the vectors of Appendix C.**

E.1 Hash and domain separation

R-40 (MUST). All commitments use **SHA-256**. Every hash input begins with a domain tag or a single-byte prefix, so that a value hashed in one role can never be reinterpreted in another. The tree prefixes are:

0x01 – internal node

0x02 – empty tree, followed by the constant **MIZAN_EMPTY**

0x03 – data leaf (OHLC bar, format v2)

Named domain tags are used where a structure is not a bar tree, for example the trial-ledger leaf domain **MIZAN_TRIAL_LEAF_V2**. **A tag is versioned, and a changed encoding takes a new tag rather than reusing the old one**, so a commitment made under one encoding can never validate under another.

E.2 Data leaves

R-40 (MUST). A data leaf binds the bar's **position** and its **full OHLC**, each as little-endian bytes, in this order:

$\text{leaf} = \text{SHA-256}(0x03 \parallel \text{index}_{u64} \parallel \text{timestamp}_{i64} \parallel \text{open}_{i64} \parallel \text{high}_{i64} \parallel \text{low}_{i64} \parallel \text{close}_{i64})$

Binding the index means bars cannot be reordered without changing the root. Binding high and low means **intraday extremes are tamper-bound exactly as closes are**, so a prover cannot shave a wick to soften a drawdown that the gate marks against. An implementation that commits only closes is not conforming, and the difference is not cosmetic: it is the difference between a drawdown that is checked and one that is asserted.

E.3 Tree construction

R-40 (MUST). Internal nodes are $\text{SHA-256}(0x01 \parallel \text{left} \parallel \text{right})$. Where a level has an odd number of nodes, the unpaired node is **promoted unchanged to the next level**; it **MUST NOT** be hashed against a copy of itself. Duplicating the final node admits distinct leaf sets that produce identical roots. This is the malleability class recorded as CVE-2012-2459, and an implementation that duplicates is unsound regardless of what else it does correctly. The empty tree hashes to $\text{SHA-256}(0x02 \parallel \text{MIZAN_EMPTY})$.

E.4 Trial-ledger leaves

R-40 (MUST). A trial leaf binds its sequence position, the commitment to the trial's specification, its registration time, and its resolved result:

```
leaf = SHA-256( MIZAN_TRIAL_LEAF_V2 || sequ64 || spec_commitment32B || registered_ati64 ||  
resolved_sharpei64, ×106 )
```

Because *seq* and the result are both bound, **trials can be neither reordered nor re-scored after commitment** without changing the root. That is the property that makes *N* structural instead of reported. An unresolved trial carries a reserved sentinel in the result field; it counts toward *N* and is excluded from the dispersion of R-38.

Prove the edge. Never reveal the strategy.

VTR-1 · THE VERIFIABLE TRACK RECORD STANDARD · VERSION 1.1 · AUGUST 2026 · © 2026 MIZAN
VERIFICATION, INC. (DELAWARE, USA) · CC BY 4.0

CANONICAL TEXT: MIZAN.MARKET/STANDARD-SPEC · REFERENCE IMPLEMENTATION & REGISTRY:

MIZAN.MARKET · THIS RENDERING IS TIMESTAMPED VIA OPENTIMESTAMPS